

Multiple Choice (10 marks)

1. Which of the following is NOT a reasonable justification for choosing to busy-wait on an asynchronous event?
 - (a) The wait is expected to be short.
 - (b) A busy-wait loop is easier to code than an interrupt handler.
 - (c) There is no other work for the processor to do.
 - (d) The program executes on a time-sharing system.
2. Which of the following sets of bit strings CANNOT be described with a regular expression?
 - (a) All bit strings whose number of zeros is a multiple of five
 - (b) All bit strings starting with a zero and ending with a one
 - (c) All bit strings with an even number of zeros
 - (d) All bit strings with more ones than zeros
3. Which of the following are most vulnerable to injection attacks?
 - (a) Session IDs
 - (b) Registry keys
 - (c) Network communications
 - (d) SQL queries based on user input
4. You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?
 - (a) No, I cannot compute the key.
 - (b) Yes, the key is $k = m \text{ xor } c$.
 - (c) I can only compute half the bits of the key.
 - (d) Yes, the key is $k = m \text{ xor } m$.
5. A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
 - (a) front-door
 - (b) backdoor
 - (c) clickjacking
 - (d) key-logging

True / False (10 marks)

Answer True or False

6. True or False: Mutation-based fuzzing exclusively applies to file-based inputs and is not suitable for network protocol testing.
7. True or False: The decimal number 0.1 can be represented exactly in binary notation without approximation.
8. True or False: WPA 2 is a recent protocol that simplifies the process of connecting devices to wireless networks.
9. True or False: Encryption and decryption provide both confidentiality and integrity by ensuring that data is secure and unaltered.
10. True or False: A packet filter firewall functions at the application or transport layer, analyzing data payloads for filtering.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether the given number is a perfect square or not.
12. Write a python function to find the minimum operations required to make two numbers equal.

End of Paper